

6	(a)	good attempt at evenly spaced parallel lines, and downward arrow.	1 1
	(b)	To produce a fine/ narrow beam /ions parallel to each other	1
	(c)	(i) <i>(region R) into page</i> <i>(region S) out of page</i>	1 1
		(ii) Electric force is downward with the magnetic force is directed upward. Electric force is equal to qE while magnetic force is equal to Bqv . They are equal to each other which gives $v = E/B$ and mass is not included in the relation.	1 1 1
		(iii) It will hit below the slit nearer to the bottom plate. Since electric force is greater than the magnetic force.	1 1
	(d)	(i) There is a change in magnetic flux linkage though the coil as the cart approaches and leaves the coil.	1
	(d)	(ii) The direction of the induced e.m.f. is such that the current would flow to oppose	1

			the change producing it. As the cart passes A, the flux linkage is increasing and as the cart passes B, magnetic flux linkage is decreasing so e.m.f. induced is in opposite direction.	1
	(d)	(iii)	The cart has accelerated between A and C/ moves faster/ more KE at C. The change of magnetic flux linkage takes place within a shorter time frame, hence rate of change of magnetic flux is greater. This induced e.m.f. is greater.	1
7	(a)	(i)	effective direct voltage that produces the same heating effect as the alternating	1